

Determining Static and Dynamic Characteristics of a Flash ADC

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Abstract

This research focuses on the static and dynamic characterization of an 8-bit Flash ADC embedded within the Secure CMOS Image Sensor (SCIS), a custom chip designed at Worcester Polytechnic Institute. Using the IEEE 1241-2023 standard as a reference, key ADC performance metrics such as offset error, gain error, DNL, INL, SNR, and SINAD were explored through Cadence simulations and analyzed using Python. This work establishes a simulation-based baseline for evaluating the fabricated chip, which will return from TSMC in September 2025. A real-world test platform will be developed using a ZTEX FPGA and EZ-USB FX3 interface, and future measurements will validate the simulation results as part of continued thesis work.

I. Introduction

This research focuses on the characterization of an internal Flash ADC of a CMOS Image Sensor. The Secure CMOS Image Sensor (SCIS) was designed by a group of undergraduate students at Worcester Polytechnic Institute: Hubert Liu, Kelu Liu, Philip Miu, and myself under the guidance of Dr. Ay and Ibrahim Bozyel (Liu et al., 2025).

Figure 1 shows a block diagram of the SCIS. An external FPGA sends an 8-bit address signal into the row/column decoders. This selects a specific pixel of the 256x420 Active Pixel Sensor (APS) (Nixon et al., 1995) (Fossum, 1993) which is then read and reset by the row drivers. The top and bottom column Analog Signal Processor (ASP) corresponds to every other column of the APS array; so, the odd columns are sent to the top ASP, and the even columns are sent to the bottom ASP. The signal from the column ASP's is then sent to the global ASP which amplifies the analog signal sent to the 8-bit Flash Analog-to-Digital Converter (ADC). The parallel 8-bit output is then encrypted using an externally programmed 16-bit key that is XOR'd to the signal, 8-bits at a time, shifting 1-bit over at each clock cycle, creating a new key each time. Finally, this encrypted signal is then serialized and sent to the FPGA (Liu et al., 2025).

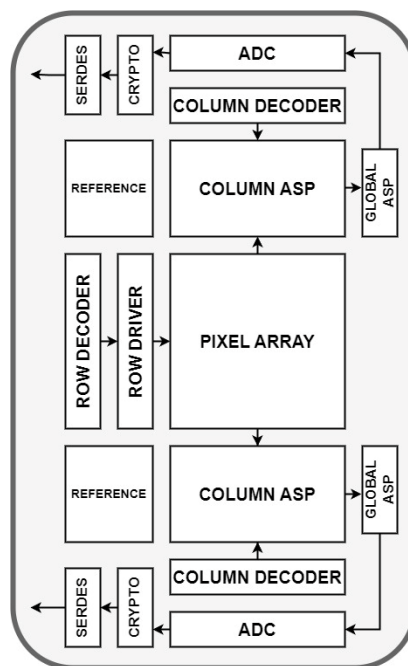


Figure 1: SCIS Block Diagram

The SCIS has a total of 3 pins dedicated to testing the pixel array and ADC, `adc_test_en`, `test_enable`, `Vtest_in`, along with the 5 pins used to deliver the scan chain (`scan_in`, `scan_clk`, `scan_rstb`, `scan_sel`, and `scan_out`), which allows for extra fine tuning of current bias's and op-amp gains throughout the different modules of the chip. An input signal like ramp or sine wave can be sent directly to the input of the FADC when `adc_test_en` is high, and we can look at the output pins of each FADC (`data_out0` and `data_out1`) for a serialized 8-bit digital output. As

mentioned earlier, there is an encryption module with a key set externally, so if the key is set to all zeros, we can analyze the unencrypted serial output directly from the chip.

This report documents the work done during my directed research during the summer 2025 semester. The goal of this research was to explore the techniques and perform tests of static and dynamic ADC characteristics on the SCIS's FADC. Section II will discuss helpful and relevant literature. Section III will discuss the static and dynamic characterization results from Cadence simulations. Lastly, section IV will discuss the planned testing setup when the chip is taped out in September 2025.

II. Literature Review

The IEEE 1241-2023 Standard for Terminology and Test Methods for Analog-to-Digital Converters provides the most up to date information on how to characterize an ADC. The standard separates the characterization methods into two types, static and dynamic. As the standard states, static errors are found by injecting a low frequency signal (like a linear ramp from 1V to 2V over 50us) and can be characterized by the non-idealities between code transition levels. Dynamic errors are found by injecting a high frequency signal (like a 500kHz sine wave) and can be characterized by the non-idealities in the frequency domain (*IEEE Standard for Terminology and Test Methods for Analog-to-Digital Converters*, n.d.). The characteristics that this report will discuss are listed in Table 1 with their definitions from the IEEE 1241-2023 standard.

Table 1: List of static and dynamic characteristics

Type	Name	Abbr.	Definition
Static	Integral Non-Linearity (LSB)	INL	The maximum and minimum values of the difference between real code transition levels (CTL) and ideal CTL's after gain and offset adjustment.
	Derivative Non-Linearity (LSB)	DNL	Maximum and minimum widths of each CTL.
	Offset Error (LSB)	--	The horizontal shift between the actual and ideal transfer curve at the low end of the ADC input range.
	Gain Error (LSB)	--	The deviation in slope between the actual and ideal transfer curve after correcting for offset.
Dynamic	Signal to Noise Ratio (dB)	SNR	$\text{RMS}(\text{SIGNAL}) / \text{RMS}(\text{NOISE})$
	Signal to Noise and Distortion Ratio (dB)	SINAD	$\text{RMS}(\text{SIGNAL}) / \text{RMS}(\text{Noise and distortion})$
	Effective Number of Bits (bits)	ENOB	The actual resolution of the ADC in bits due to SINAD
	Total Harmonic Distortion (dBFS)	THD	THD is estimated by the rss of the second through the tenth harmonics (including alias's), inclusive. (IEEE 1241)
	Spurious Free Dynamic Range (dBFS)	SFDR	The magnitude of the first harmonic in dBFS
	Dynamic Range (dBFS)	DR	The magnitude of the last harmonic in dBFS

When an imager has an integrated ADC, it is important to be able to characterize the ADC with the errors above in order to compensate for characteristics like gain and offset error. Without direct access to the input of the FADC, the only other way to test it would be through the pixel array. This technique is not advised because it would produce results that are a combination of both the pixel characteristics and ADC characteristics. Morishita et al. (2020) provides a sample test setup for characterizing a single-slope column parallel ADC using an optical input as their input signal. The optical source produced different patterns that could be used to measure noise errors, static characteristics, or column interference. The only test that could be done without worrying about the quality of the optical source was the noise errors. The column interference test was impossible because the pattern of the source could never isolate one column at a time. Their solution was to have a set of internal switches and a multiplexer that could select between different external electrical inputs. Rather than a multiplexer, the SCIS uses the enable pins to switch the input of the ADC to either the pixel array or the testing signal from a function generator. Furthermore, the bottom of the pixel array has a single row of pixels dedicated to testing the column interference. This design choice ensures the ADC can be properly characterized independent of pixel behavior, aligning with best practices for accurate, isolated testing as demonstrated in prior literature.

In addition to isolating the ADC, the quality of the clock signals that control the chip is another important factor that will affect the results of static and dynamic tests. There are multiple clocks that will control the SCIS from an FPGA board, and each of these clocks suffer from non-idealities such as aperture jitter and clock jitter. S. Chaganti et al. (2017) presents a method of estimating the clock jitter of an ADC with a test setup that could potentially be recreated using the SCIS chip. Their method requires two ADC's running on the same clock and input signal. After acquiring two sets of coherently sampled output data for each ADC, the true specifications can be estimated by calculating the clock jitter variance. They found that this method could tolerate a clock jitter of up to 5 to 6 times the aperture jitter. In a future publication, S. K. Chaganti et al. (2018) provides a method of breaking up a larger sample of data that does require coherent sampling. These methods could possibly be used later in the year if it seems that there is too much spectral leakage or noise due to the quality of equipment that is available.

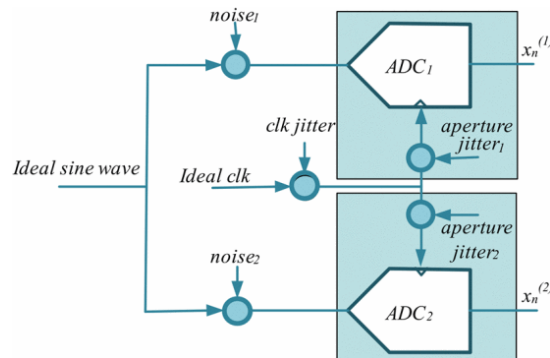


Figure 2: Test setup from S. Chaganti et al. (2017)

III. Simulations

The following simulations were all done in Cadence IC Design and the calculated errors and plots were generated using Python and can be found on GitHub (<https://github.com/Vision314/ADC-Static-and-Dynamic-Characterization>). Table 2 shows the results of all the simulations discussed below.

Table 2: Static and dynamic ADC characteristics of the SCIS FADC

Type	Name	Definition
Static	INL	+12.65/-1.01
	DNL	+1.816/-0.744
	Gain	1.09 LSB
	Offset	-2.86 LSB
Dynamic	SNR	44.55 dB
	SINAD	44.45 dB
	ENOB	7.39 bits
	THD	-29.17 dBFS
	SFDR	33.52 dBFS
	DR	46.50 dBFS

Static Characterization

Figure 3 shows the test setup to calculate static errors. The input signal is a linear ramp from 1V to 2V (Full Scale) over 52.88 μ s. This is the amount of time it takes to get 256 samples at a clock rate of about 5MHz.

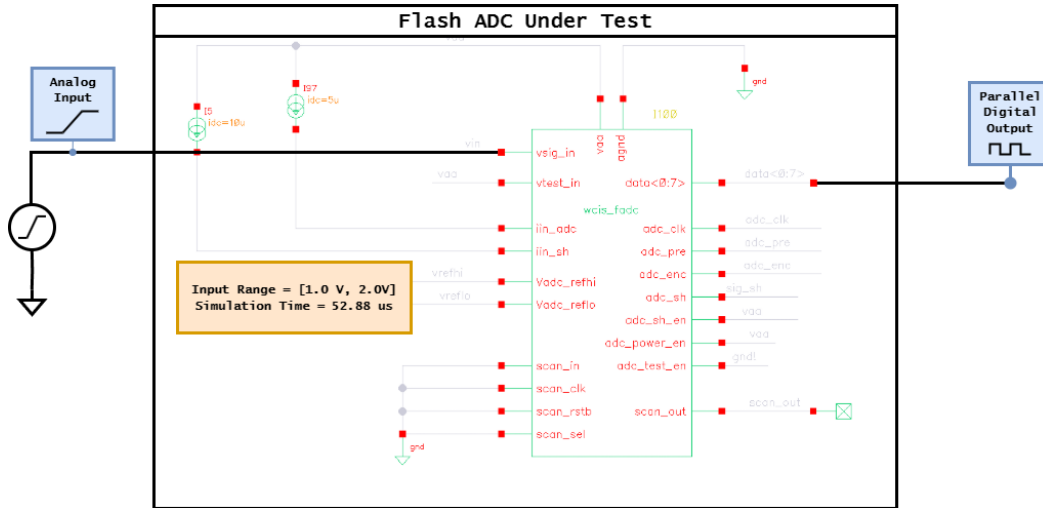


Figure 3: Static characterization test setup

Figure 4(a) shows the CTLs directly from Cadence (red) compared to the ideal CTLs (green). The red line is not easy to calculate errors on so we need to clean it up by getting the corners of each step to calculate offset, gain, and DNL. First, we remove duplicate or non-increasing input values, then scan backward through the signal to capture only the points where

the output code changes, filtering out voltage spikes and isolating the true code transition levels. What we end up getting is a line that is just made up of each corner step of the CTL's from cadence. This can be seen as the red dots in Figure 4 (c) and (d).

Next we can calculate the offset and gain error. For this test we will be using the end-to-end fit which matches the start and end of the non-ideal CTL's to the start and end of the ideal CTL's, leaving a gap in the middle. The offset error is how much we have to shift the non-ideal CTL's to match the ideal CTL's. This can be seen in Figure 4 (c). The gain error is how much we have to scale the non-ideal CTL's to match the ideal CTL's. This can be seen in Figure 4 (d). The results of this gain and offset adjustment can be seen in Figure 4 (b).

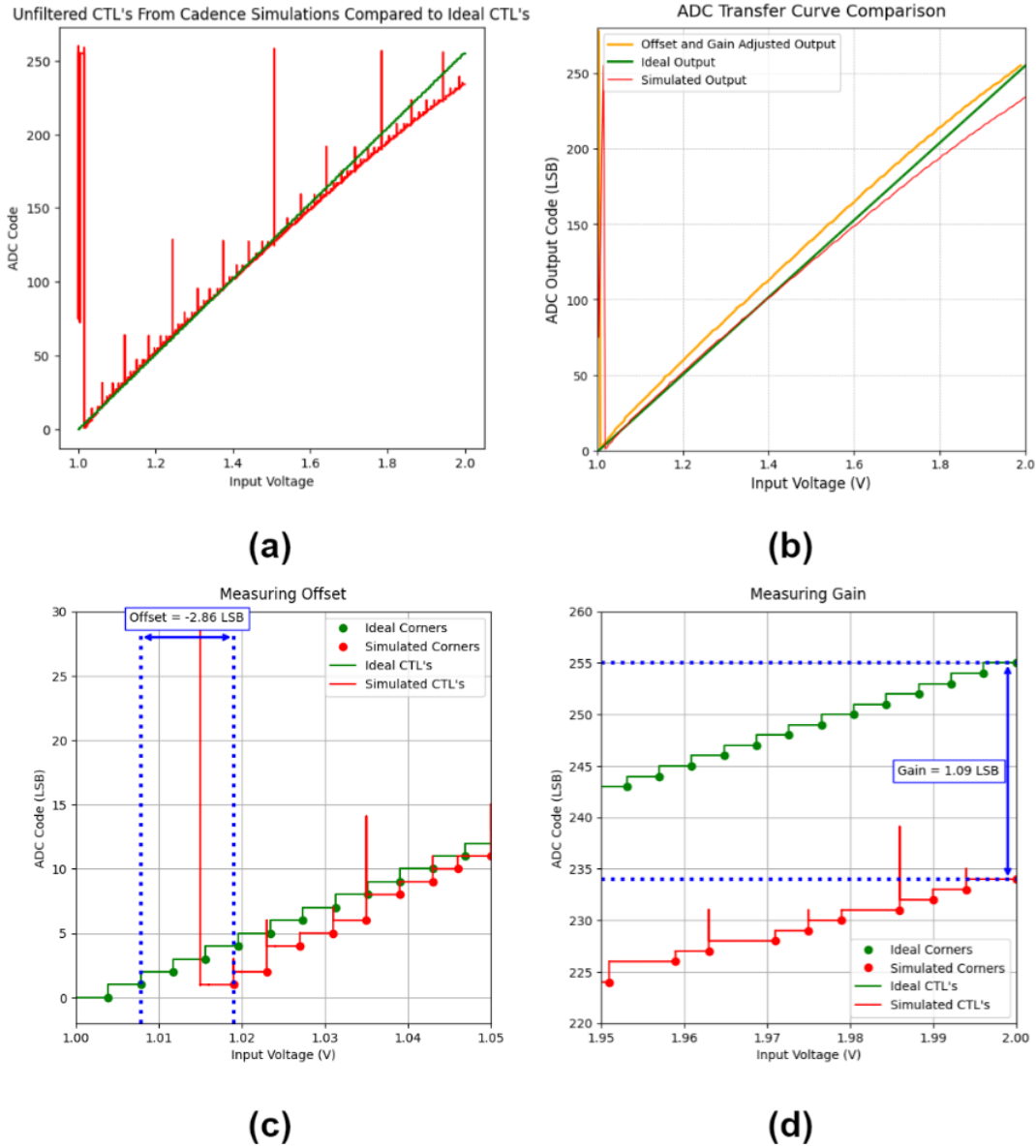


Figure 4: (a) CTL's directly from cadence vs ideal CTL's (b) Simulated, ideal, and adjusted CTL's (c) Offset calculation (d) Gain calculation

The last two metrics for static characterization are derivative non-linearity (DNL) and integral non-linearity (INL). DNL measures the deviation of each code width from the ideal 1 LSB step, as shown in Figure 5(a), where shorter or longer steps create negative or positive DNL, respectively. Figure 5(b) displays the calculated DNL across all codes, with a worst-case value of +1.816 LSB and a minimum of -0.744 LSB, indicating no missing codes. INL, shown in Figures 5(c) and 5(d), takes the midpoints of each step and represents the deviation of actual CTL from the ideal CTL, reaching a maximum deviation of +12.65 LSB.

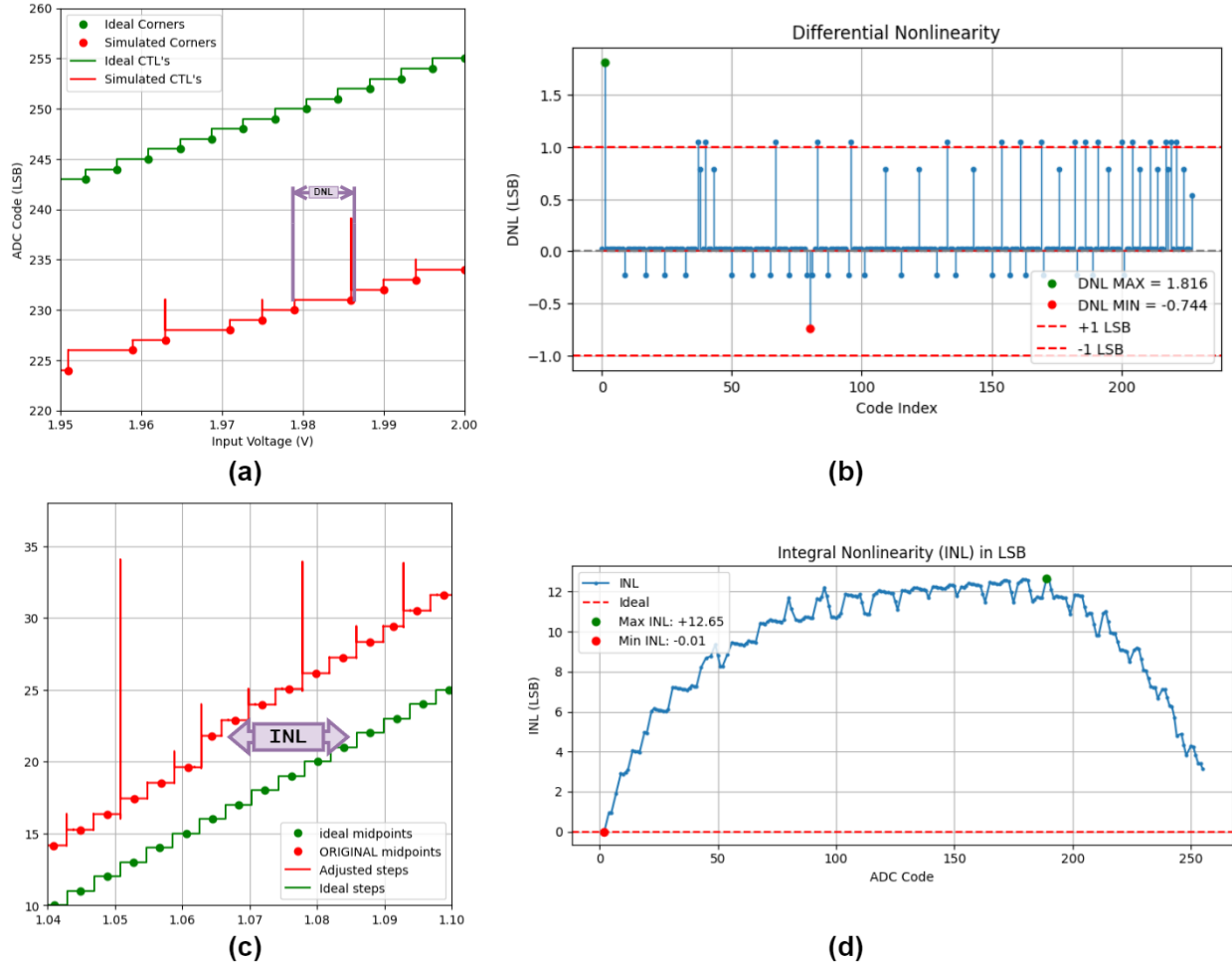


Figure 5: (a) Method for calculating DNL (b) DNL results (+1.816/-0.744) (c) Method for calculating INL (d) INL results (+12.65/-1.01)

Dynamic Characterization

Figure 7 shows the test setup to acquire all the dynamic test results. When we apply an analog signal to the FADC it outputs 8 parallel digital outputs. Using an ideal DAC in Cadence, we can reconstruct the original signal.

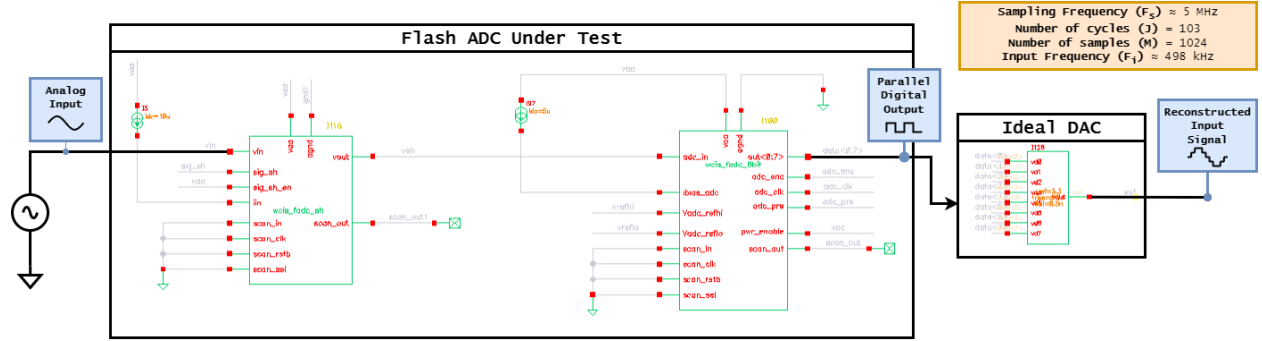


Figure 6: Dynamic characteristics test setup

From the IEEE 1241 standard, the equation for calculating the input frequency is shown below:

$$MF_i = JF_s \quad (1)$$

where

- F_s is the sampling frequency
- J is the total integer number of cycles
- F_i is the frequency of the input signal
- M is the total number of samples

The selected sampling period (T_s) is 200ns with a rise and fall time of 1ns so the sampling frequency ($F_s = 1/T_s = 1/202\text{ns}$) is 4950495.0495 Hz (~5Mhz). The number of samples (M) determines the resolution of the spectrum plot, a common number of samples is 1024 (Lotfi et al., 2023; Mielke, 1996). To select the integer number of cycles (J) and input frequency (F_i), two conditions must be met: (1) the sampling frequency must exceed twice the input frequency (Nyquist theorem), and (2) the ratio M/J must be irreducible ($J = 103$). Dividing M by 10 (to comfortably exceed Nyquist) and choosing the nearest prime number ensures both conditions are satisfied. Finally, F_i can be calculated using Equation (1).

After the simulation is run in Cadence, the data can be sent to a CSV file and analyzed in Python. Figure 8 (a) shows the results of the transient simulation after being centered. Figure 8 (b) shows the positive frequencies of the FFT after normalizing to the number of samples. Figure 8 (c) shows the FFT converted to dBFS (decibels referenced to full scale).

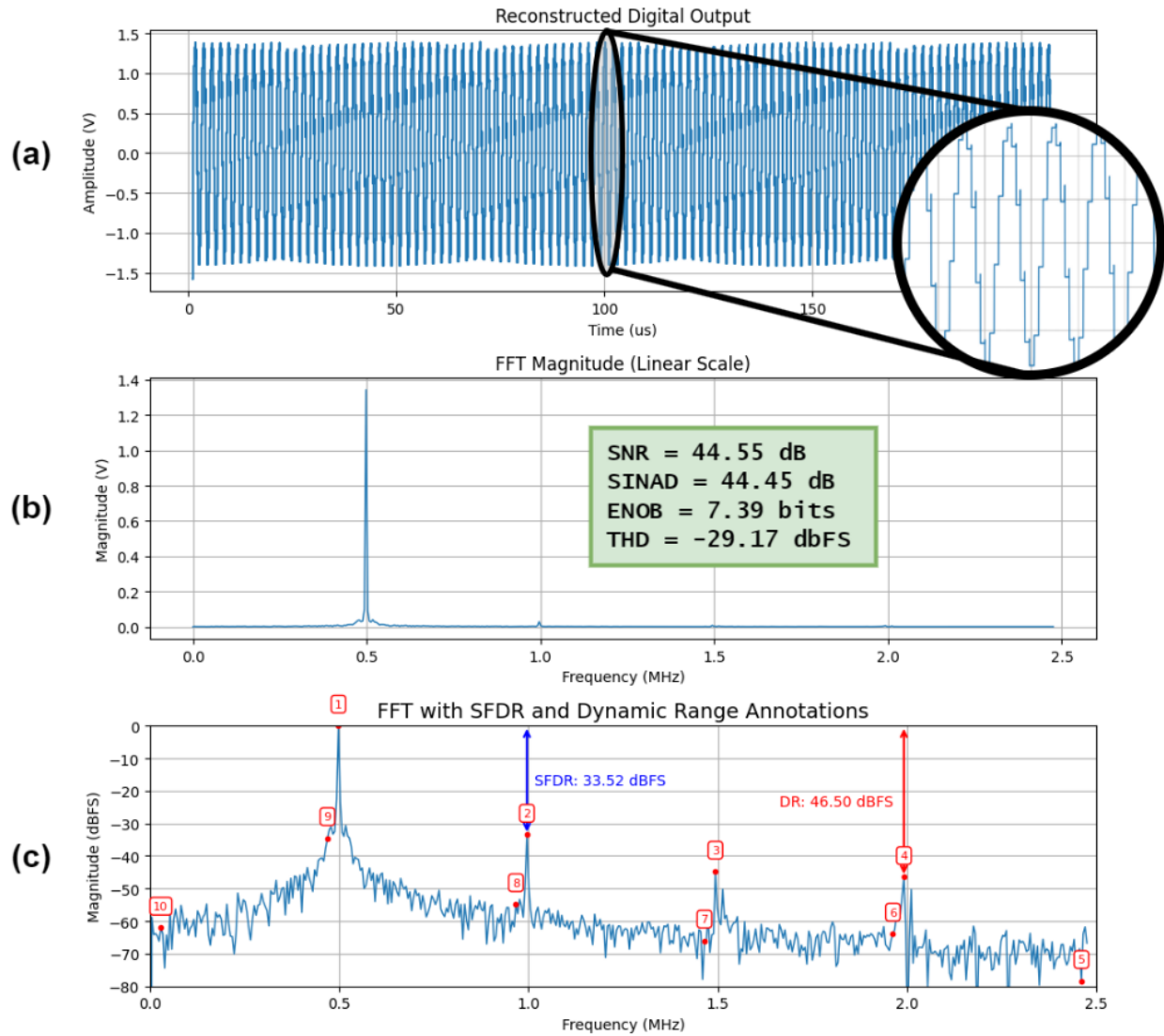


Figure 7: Dynamic characteristics results (a) Transient response (b) FFT in (V) (c) FFT in dBFS

IV. Real-World Testing Setup for SCIS

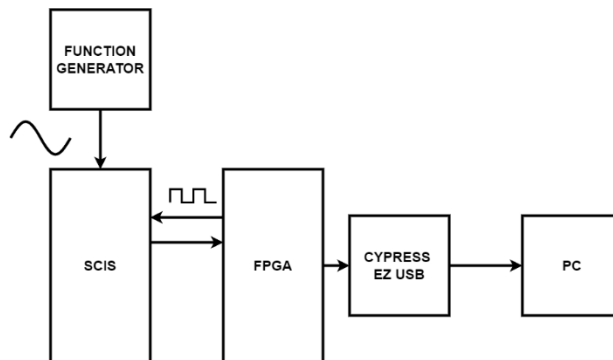


Figure 8: Real test setup for the SCIS ADC performance

When the chip returns from TSMC in September 2025, I will begin testing and debugging using the ZTEX 2.14b FPGA board (Artix-7), which interfaces with the EZ-USB FX3. This board will generate all necessary clock signals and read the ADC outputs. As shown in Figure 9, a function generator will supply either a sine or ramp input for dynamic or static tests, respectively. The captured data will be processed using the Python scripts used above to extract ADC performance metrics.

V. Conclusion

This directed research served as an important steppingstone in preparing for the full characterization of the SCIS chip once it is returned from TSMC in September 2025. Through literature review, Cadence simulations, and Python-based analysis, I developed a much clearer understanding of how to properly test and evaluate the performance of a Flash ADC embedded within a CMOS image sensor. Prior to this work, my understanding of ADC characterization, particularly the physical process to determine the errors, was uncertain. Now, I have a solid foundation to build upon when setting up real hardware experiments in the fall. The methods and test procedures documented in this report will guide the lab setup and measurements during my thesis work. I'm excited to see how the simulated results compare to actual measurements and to begin the next stage of testing on the fabricated SCIS.

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